

# Vector Integration: Work Done Around a Closed Elliptical Path

## 1. Work Done and Path Parameterization for an Ellipse

### Name of Concepts

- Work Done by a Force Field along a Closed Loop.
- Parametric Representation of a 2D Ellipse ( $z = 0$ ).
- Conservative Field Validation.

### Core Mathematical Formulas

The total work done by a force field  $\vec{F}$  moving a particle along a path  $C$  is:

$$\text{Work Done} = \oint_C \vec{F} \cdot d\vec{r} = \oint_C (f_1 dx + f_2 dy + f_3 dz)$$

The parametric equations for an ellipse centered at the origin with semi-major axis  $a$  and semi-minor axis  $b$  are:

$$x = a \cos \theta, \quad y = b \sin \theta$$

To map out the entire elliptical boundary exactly once in the counter-clockwise direction, the angular parameter  $\theta$  spans the interval:

$$0 \leq \theta \leq 2\pi$$

### Beginner-Friendly Explanation of Operations

- Closed-Loop Line Integral ( $\oint$ ):** The circle on the integration symbol indicates that our path forms a complete, unbroken loop starting and ending at the same location.
- Ellipse Parameterization:** Just like a circle of radius  $r$  uses  $x = r \cos \theta$  and  $y = r \sin \theta$ , an ellipse stretches differently along its two axes. We replace the single radius  $r$  with the independent horizontal stretch factor  $a$  and vertical stretch factor  $b$ . This allows us to sweep along the entire curve by tracking a single angle variable  $\theta$ .
- Flat 2D Layer ( $z = 0$ ):** Since the problem locks the path to the  $z = 0$  plane, the particle never changes its vertical altitude. Therefore, the microscopic vertical step size  $dz$  is exactly 0. Any force component pushing up or down along the  $\vec{k}$  axis does zero work and can be completely omitted from the calculation.
- Testing for Conservative Nature:** A field is conservative if its curl evaluates to zero ( $\nabla \times \vec{F} = \vec{0}$ ). If a field *is* conservative, the total work done along *any* closed loop must equal exactly 0 because the starting and ending points are identical. If our final work calculation yields a non-zero number, it serves as immediate proof that the field is non-conservative.

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## 2. Step-by-Step Problem Solution

### Question

Find the work done in moving a particle once around the ellipse  $\frac{x^2}{25} + \frac{y^2}{16} = 1$ ,  $z = 0$  under the field of force given by  $\vec{F} = (2x - y + z)\vec{i} + (x + y - z^2)\vec{j} + (3x - 2y + 4z)\vec{k}$ . Is the field conservative?

## Step 1: Extract the geometric values of the ellipse

Match the given ellipse equation to the standard form  $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ :

$$a^2 = 25 \implies a = 5$$

$$b^2 = 16 \implies b = 4$$

## Step 2: Establish the parametric coordinate equations

Substitute the values of  $a$  and  $b$  into the trigonometric parametric layout. Account for the path constraint  $z = 0$ :

$$x = 5 \cos \theta$$

$$y = 4 \sin \theta$$

$$z = 0$$

## Step 3: Compute the differentials in terms of $d\theta$

Differentiate each coordinate function with respect to  $\theta$ :

$$\frac{dx}{d\theta} = -5 \sin \theta \implies dx = -5 \sin \theta d\theta$$

$$\frac{dy}{d\theta} = 4 \cos \theta \implies dy = 4 \cos \theta d\theta$$

$$\frac{dz}{d\theta} = 0 \implies dz = 0$$

## Step 4: Simplify the field components using the $z = 0$ constraint

Extract the scalar components of  $\vec{F}$  and substitute  $z = 0$  to simplify them before integration:

$$f_1 = 2x - y + z = 2x - y + 0 = 2x - y$$

$$f_2 = x + y - z^2 = x + y - 0^2 = x + y$$

$$f_3 = 3x - 2y + 4z = 3x - 2y + 0 = 3x - 2y$$

## Step 5: Substitute coordinates into the active components

Express  $f_1$  and  $f_2$  purely as functions of our angle variable  $\theta$ :

$$f_1 = 2(5 \cos \theta) - (4 \sin \theta) = 10 \cos \theta - 4 \sin \theta$$

$$f_2 = (5 \cos \theta) + (4 \sin \theta) = 5 \cos \theta + 4 \sin \theta$$

## Step 6: Set up the scalar work integral equation

Substitute our simplified functions and differentials into the line integral equation, setting the boundary limits from  $\theta = 0$  to  $2\pi$ :

$$\text{Work Done} = \int_0^{2\pi} [(f_1)dx + (f_2)dy + (f_3)dz]$$

Since  $dz = 0$ , the third component drops out completely:

$$\text{Work Done} = \int_0^{2\pi} [(10 \cos \theta - 4 \sin \theta)(-5 \sin \theta d\theta) + (5 \cos \theta + 4 \sin \theta)(4 \cos \theta d\theta)]$$

## Step 7: Distribute and expand terms inside the integrand

Factor out the common differential operator  $d\theta$  and multiply the terms across the parentheses:

$$\text{Work Done} = \int_0^{2\pi} [-50 \sin \theta \cos \theta + 20 \sin^2 \theta + 20 \cos^2 \theta + 16 \sin \theta \cos \theta] d\theta$$

Combine the matching trigonometric terms:

$$-50 \sin \theta \cos \theta + 16 \sin \theta \cos \theta = -34 \sin \theta \cos \theta$$

Factor out the common coefficient from the squared terms:

$$20 \sin^2 \theta + 20 \cos^2 \theta = 20(\sin^2 \theta + \cos^2 \theta)$$

Apply the fundamental trigonometric identity  $\sin^2 \theta + \cos^2 \theta = 1$ :

$$20(1) = 20$$

Assemble the simplified expression:

$$\text{Work Done} = \int_0^{2\pi} [20 - 34 \sin \theta \cos \theta] d\theta$$

## Step 8: Apply trigonometric identities to ease integration

Use the double-angle identity  $2 \sin \theta \cos \theta = \sin(2\theta)$  to simplify the second term:

$$34 \sin \theta \cos \theta = 17(2 \sin \theta \cos \theta) = 17 \sin(2\theta)$$

Substitute this back into the definite integral:

$$\text{Work Done} = \int_0^{2\pi} [20 - 17 \sin(2\theta)] d\theta$$

## Step 9: Integrate and evaluate across boundaries

Apply standard calculus integration rules ( $\int \sin(2\theta) d\theta = \frac{-\cos(2\theta)}{2}$ ):

$$\text{Work Done} = \left[ 20\theta - 17 \left( \frac{-\cos(2\theta)}{2} \right) \right]_0^{2\pi}$$

$$\text{Work Done} = \left[ 20\theta + \frac{17}{2} \cos(2\theta) \right]_0^{2\pi}$$

Evaluate the expression by substituting the upper limit ( $2\pi$ ) and subtracting the lower limit (0):

$$\text{Work Done} = \left( 20(2\pi) + \frac{17}{2} \cos(4\pi) \right) - \left( 20(0) + \frac{17}{2} \cos(0) \right)$$

Since  $\cos(4\pi) = 1$  and  $\cos(0) = 1$ :

$$\text{Work Done} = \left( 40\pi + \frac{17}{2}(1) \right) - \left( 0 + \frac{17}{2}(1) \right)$$

$$\text{Work Done} = 40\pi + \frac{17}{2} - \frac{17}{2} = 40\pi$$

## Step 10: Determine field conservative status

A force field is conservative if and only if the total work executed along a closed loop path equals exactly zero.

Because the calculated work done is  $40\pi \neq 0$ , the vector field  $\vec{F}$  is **not conservative**.